

Is there a relationship between voter turnout and county urbanness?

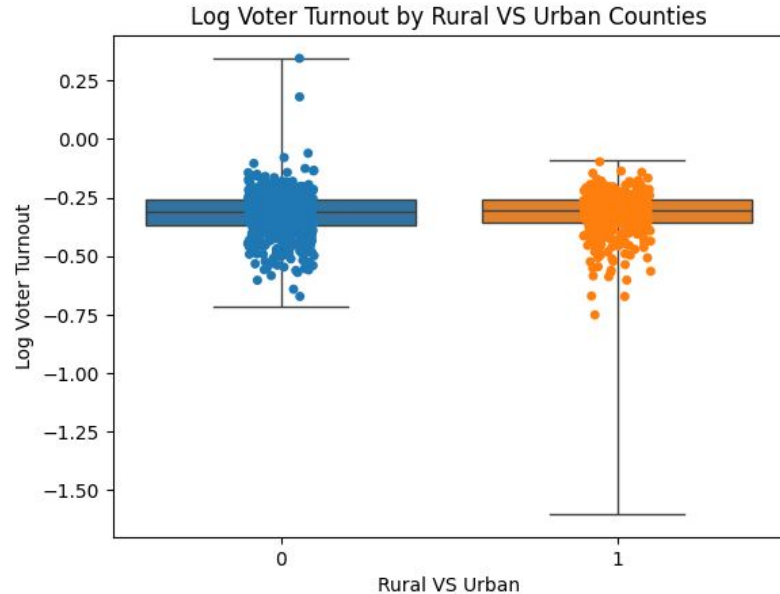
Castner, Vaughan

Null Hypothesis (H0):

The urbanness of a county has no impact on its voter turnout.

Alternative Hypothesis (HA):

The urbanness of a county has an impact on its voter turnout.



$$\log_voterturnout = \beta_0 + \beta_1 * (\text{rural_urban}) + \beta_2 * (\text{personalincomethousand}) + \varepsilon$$

A relationship exists between county urbanness and voter turnout.

Rural/urban coefficient: -0.127; p-value = 0.001

Income coefficient: 0.0012, p-value = 0.000

- OLS assumption of equal variance violated, with residual plot showing heteroskedasticity.

- Conclusion: Voter turnout decreases from rural to urban areas when controlling for income by 0.127 points.

